

# Chapter 9: Gases & Kinetic Molecular Theory

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CHM111

Lecture 9



# Part I

## Gas Pressure

# Outline of Part I: Gas Pressure

Introduction to Gases

Gas Pressure

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Introduction to Gases

Gas Pressure

# Properties of Gases

## What Makes Gases Unique?

- ▶ Gases **expand to fill** any container they occupy.
- ▶ Gases are highly **compressible**: volume changes substantially with pressure.
- ▶ Gases have much **lower density** than solids or liquids.
- ▶ Gases and liquids are both **fluids**: they flow and take the shape of their container.



### Gases = Compressible

1. Gas molecules are widely separated.
2. Most of a gas's volume is empty space.
3. Compression removes that empty space.
4.  $\therefore$  gases compress easily.



### Liquids $\approx$ Incompressible

1. Liquid molecules are already close together  
 $\Rightarrow$  little empty space to eliminate during compression.
2. Further compression pushes molecules into the steeply repulsive region of their intermolecular potential.
3. Very large pressures are therefore required for small decreases in volume.
4.  $\therefore$  liquids are only slightly compressible.

# Molecular Compression

## ▶ Question

- ▶ How do we know that *empty space* compresses, not the *molecules* themselves?

## ▶ Evidence

- ▶ Gases compress; liquids and solids scarcely do.
- ▶ Scattering shows condensed particles nearly touch.
- ▶ Gas molecules have much larger separations.

## ▶ Reasoning

- ▶  $\therefore$  condensed particles (in solids and liquids) already nearly touch
- ▶  $\therefore$  gas compression removes mostly *empty space*.

## ▶ Takeaway

- ▶ Gas compressibility reveals widely separated molecules.

# Why do Gases and Liquids Flow, but Solids do not?

## Question

- ▶ Why do gases and liquids flow, rather than solids?  
What structural feature makes flow possible?

## Evidence

- ▶ Liquid particles remain close yet continuously exchange neighbors.
- ▶ Gas particles are widely separated and move freely.
- ▶ Solid particles mainly vibrate about fixed positions.

## Reasoning

- ▶  $\therefore$  Liquids flow despite high density, spacing alone cannot explain flow.
- ▶  $\therefore$  Flow requires *particle mobility*, not merely particle separation.

## Takeaway

- ▶ Flow depends on particles changing relative positions.

# Outline of Part I: Gas Pressure

Introduction to Gases

Gas Pressure



# Pressure

Definition: Pressure ( $P$ ) — SI Unit: pascal (Pa)

Pressure is the force exerted per unit area on a surface:

$$P = \frac{F}{A}$$

where  $F$  is force (N) and  $A$  is area ( $\text{m}^2$ ).

- ▶ **Atmospheric pressure** arises from the weight of the air above Earth's surface.
- ▶ **Gas pressure** in a container results from molecules colliding with the container walls.

# Why Does a Gas Exert Pressure?

## Question

- ▶ If a gas is "mostly empty space," then why does it still exert a well-defined pressure on the container walls at every instant?

## Evidence

- ▶ Gas particles move randomly throughout the container.
- ▶ Collisions with the walls transfer *momentum*.
- ▶ Enormous particle numbers produce countless collisions each second.

## Reasoning

- ▶  $\therefore$  Frequent momentum transfers outweigh empty space, continuous contact is unnecessary.
- ▶  $\therefore$  *Pressure* arises from the cumulative effect of wall collisions.

## Takeaway

- ▶ Pressure emerges from countless particle-wall collisions.

# Why does Pressure not have direction?

## ▶ Question

- ▶ Why does *pressure* have no direction although molecular momentum is directional?

## ▶ Evidence

- ▶ Molecules transfer vector momentum during wall collisions.
- ▶ At equilibrium, molecular motion is equally likely in every direction.

## ▶ Reasoning

- ▶  $\therefore$  Opposite momentum directions occur equally, directional effects cancel.
- ▶  $\therefore$  Only the *normal* momentum transfer per unit area remains.
- ▶  $\therefore$  Pressure depends only on magnitude, not direction.

## ▶ Takeaway

- ▶ *Isotropy* (symmetry despite rotation) makes pressure non-directional.

## Units of Pressure

| Unit               | Symbol | Value at Standard Conditions |
|--------------------|--------|------------------------------|
| Pascal             | Pa     | 101 325 Pa                   |
| Kilopascal         | kPa    | 101.325 kPa                  |
| Atmosphere         | atm    | 1 atm (definition)           |
| Bar                | bar    | $\approx 1.013\,25$ bar      |
| Millimeters of Hg  | mmHg   | 760 mmHg                     |
| Torr               | torr   | 760 torr                     |
| Pounds per sq. in. | psi    | 14.696 psi                   |

- ▶ 1 mmHg  $\approx$  1 torr (essentially identical)
- ▶ Conversions: 1 atm = 760 torr = 101 325 Pa

## Check Your Understanding

Convert 1.50 atm to (a) torr and (b) kPa.

Come to lecture to see the answer.

# What Connects Weight of Air and Gas Pressure?

## ▶ Question

- ▶ Why can atmospheric pressure be explained by both air's weight and molecular collisions?

## ▶ Evidence

- ▶ Air molecules continually collide with every surface.
- ▶ Higher air columns contain more molecules above each unit area.
- ▶ More overlying molecules require stronger upward support.

## ▶ Reasoning

- ▶  $\therefore$  supporting more air requires more collision force, weight alone cannot act without collisions.
- ▶  $\therefore$  *atmospheric pressure* is the macroscopic effect of *molecular collisions* transmitting the air's weight.

## ▶ Takeaway

- ▶ Atmospheric pressure can be explained at two *scales*.

# Manometers

## Measuring Gas Pressure

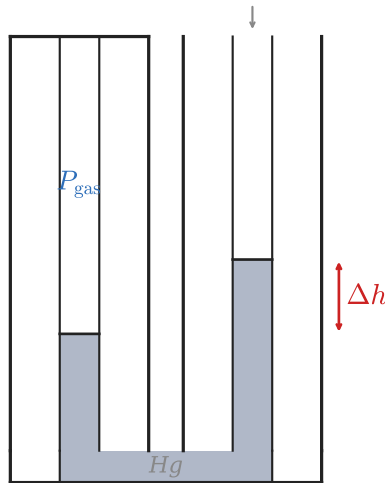
A **manometer** is a U-shaped tube containing mercury used to measure the pressure of a trapped gas.

### Open-end manometer:

- ▶ One arm connects to the gas sample.
- ▶ The other arm is open to the atmosphere.
- ▶ The mercury level difference  $\Delta h$  gives the pressure difference.

$$P_{\text{gas}} = P_{\text{atm}} \pm \Delta h \text{ (in mmHg)}$$

Use + when the gas side is *lower*; use - when the gas side is *higher*.



# Reading a Manometer

## Example

$P_{\text{atm}} = 760 \text{ mmHg}$ . The gas side of the manometer is 30 mm **lower** than the open side. What is  $P_{\text{gas}}$ ?

$$P_{\text{gas}} = P_{\text{atm}} + \Delta h$$

$$P_{\text{gas}} = 760 \text{ mmHg} + 30 \text{ mmHg}$$

$$P_{\text{gas}} = 790 \text{ mmHg}$$

- ▶ Gas side *lower*  $\Rightarrow$  gas pressure is *greater* than atmospheric.
- ▶ Gas side *higher*  $\Rightarrow$  gas pressure is *less* than atmospheric.



## Check Your Understanding

A gas sample is connected to an open-end manometer.

$$P_{\text{atm}} = 760 \text{ torr.}$$

The mercury on the **gas side** is 45 mm **higher** than the open side.

What is  $P_{\text{gas}}$ ?

Come to lecture to see the answer.

## Part II

# Gas Laws

## Outline of Part II: Gas Laws

Temperature and the Gas Variables

The Simple Gas Laws

The Ideal-Gas Law

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# Variables That Describe a Gas

The **state** of any gas sample is fully described by four variables (use either  $n$  or  $N$ ):

| Variable    | Symbol | SI Unit      | Common Units                            |
|-------------|--------|--------------|-----------------------------------------|
| Pressure    | $P$    | Pa           | atm, mmHg, torr, kPa                    |
| Volume      | $V$    | $\text{m}^3$ | L, mL                                   |
| Temperature | $T$    | K            | $^{\circ}\text{C}$ , $^{\circ}\text{F}$ |
| Amount      | $n$    | mol          | —                                       |
| # Molecules | $N$    | —            | —                                       |

Temperature **must always be in Kelvin** in gas law calculations.

$$T(\text{K}) = T(^{\circ}\text{C}) + 273.15$$

# What is Temperature?

## Temperature ( $T$ ) — SI Unit: kelvin (K)

Temperature is proportional to the **average internal kinetic energy** of the atoms and molecules that compose a substance.

$$T \propto \overline{KE}_{\text{internal}}$$

- ▶  the *average* molecule composing a high-temperature object *moves faster* than the *average* molecule composing a low-temperature object.
- ▶ Temperature is defined for a system in *thermal equilibrium*, where the molecules have a stable distribution of energies.
- ▶ *Why is Temperature an Equilibrium Property?*
  - ▶ For non-thermal-equilibrium systems, different parts of a system have different energy distributions.
  - ▶  $\therefore$  a single temperature cannot describe the entire system.

# Why Temperature Tracks Average Kinetic Energy

## Question

- ▶ Why does temperature depend on *average kinetic energy* rather than other energies?

## Evidence

- ▶ Identical samples at equal temperature can contain different numbers of particles.
- ▶ *Total energy* changes with sample size, yet temperature does not.
- ▶ Faster microscopic motion produces greater pressure during particle collisions.

## Reasoning

- ▶  $\therefore$  temperature is size-independent, total energy cannot determine it.
- ▶  $\therefore$  pressure reflects particle speeds, *kinetic energy* measures thermal motion directly.
- ▶  $\therefore$  temperature is proportional to the *average* kinetic energy per particle.

## Takeaway

- ▶ Temperature measures average microscopic motion, not total energy.

## Temperature vs. Heat

### Temperature

- ▶ A property of the object itself.
- ▶ Related to the **average** KE of molecules.
- ▶ Measured in °C, °F, or K.

### Heat

- ▶ Energy transferred between objects.
- ▶ Flows from higher  $T$  to lower  $T$ .
- ▶ Measured in joules (J).

A spark from a fire has a higher temperature than boiling water, but a pot of boiling water transfers *more heat* to your hand than the spark because it contains far more molecules.



# Why Does Heating Increase Molecular Motion?

## Question

- ▶ Why does heating increase molecular motion instead of other forms of energy?

## Evidence

- ▶ Heating transfers energy through countless molecular collisions.
- ▶ Collisions continually redistribute energy among translation, rotation, and vibration.
- ▶ At thermal equilibrium, each type of molecular motion (translation, rotation, and vibration) gains a predictable average share.

## Reasoning

- ▶  $\therefore$  collisions continually share added energy among molecular motions,
- ▶  $\therefore$  heating increases molecular *kinetic energy*, not a separate energy store.
- ▶  $\therefore$  temperature rises because average molecular motion increases.

## Takeaway

- ▶ Heating raises temperature by increasing average molecular motion.

# Why Is Temperature the Average Molecular Kinetic Energy?

## ▶ Question

- ▶ What observation forces temperature to represent *average molecular kinetic energy*?

## ▶ Evidence

- ▶ Zero net exchange of energy between two gases when both are at the same temperature.

## ▶ Reasoning

- ▶  $\therefore$  Energy flow stops only when the exchanged quantity becomes equal.
- ▶  $\therefore$  Composition differs, equal internal structure cannot explain equilibrium.
- ▶  $\therefore$  The shared quantity is *average molecular kinetic energy*, measured as *temperature*.

## ▶ Takeaway

- ▶ Temperature measures average molecular kinetic energy.

# Temperature Scales

## Fahrenheit

- ▶ Water freezes:  $32^{\circ}\text{F}$
- ▶ Water boils:  $212^{\circ}\text{F}$
- ▶ Body temp:  $98.6^{\circ}\text{F}$
- ▶ Absolute zero:  
 $-459.67^{\circ}\text{F}$

## Celsius

- ▶ Water freezes:  $0^{\circ}\text{C}$
- ▶ Water boils:  $100^{\circ}\text{C}$
- ▶ Body temp:  $37.0^{\circ}\text{C}$
- ▶ Absolute zero:  
 $-273.15^{\circ}\text{C}$

## Kelvin

- ▶ Water freezes:  $273.15\text{ K}$
- ▶ Water boils:  $373.15\text{ K}$
- ▶ Body temp:  $310.15\text{ K}$
- ▶ Absolute zero:  $0\text{ K}$

# Temperature Scales

## Converting Between Temperature Scales

$$T_K = T_C + 273.15$$

$$T_C = \frac{5}{9}(T_F - 32)$$

$$T_F = \frac{9}{5} T_C + 32$$

## Example

Convert 25.0 °C to Kelvin.

$$T_K = 25.0\text{ °C} + 273.15 = \boxed{298.15\text{ K}}$$

The Kelvin scale starts at **absolute zero** (0 K), the lowest possible temperature. There are no negative Kelvin values.

# Absolute Zero

## Absolute Zero: $T = 0 \text{ K}$

- ▶ At  $T > 0 \text{ K}$ , molecules occupy excited energy states due to thermal energy. This produces the familiar random translational, rotational, and vibrational motion.
- ▶ At  $T = 0 \text{ K}$ , those thermal excitations disappear.  $\therefore$  the system is in its quantum ground state, particles still exhibit zero-point motion.

## Absolute Zero is Not Attainable

- ▶ The Third Law of Thermodynamics prevents the attainment of  $T = 0 \text{ K}$ .

## Why Kelvin?

- ▶ Kelvin is an **absolute** scale:  $0 \text{ K}$  means zero thermal energy.
- ▶  $0^\circ\text{C}$  and  $0^\circ\text{F}$  are arbitrary reference points — they do *not* imply zero thermal energy.
- ▶ Use Kelvin in calculations. (but remember  $\Delta T_C = \Delta T_K$ )

## Why Matter must be in Motion

1. An electron in an atom must be *confined* near the nucleus ( $\therefore$  it belongs to the atom).
2. Confinement implies finite uncertainty in position ( $\Delta x \neq \infty \therefore$  we know roughly where the electron is).
3. The uncertainty principle requires a nonzero spread in momentum ( $\Delta p \neq 0 \therefore \Delta x \neq \infty$ ).
4. Uncertainty in momentum  $\Rightarrow$  nonzero momentum  $\Rightarrow$  nonzero speed  $\Rightarrow$  nonzero kinetic energy.
5.  $\therefore$  the electron must be in motion (even if  $T = 0 \text{ K}$ )

The balance between this unavoidable kinetic energy and the attractive electric potential determines the atom's stable size.

# Why Uncertainty Gives Zero-Point Kinetic Energy

## ► Question

- Why does momentum uncertainty produce nonzero kinetic energy?

## ► Evidence

- Confinement requires  $\Delta x \neq 0$ , so  $\Delta p \neq 0$ .
- Kinetic energy depends on  $\langle p^2 \rangle$ , not  $\langle p \rangle$ . — *Why does this matter?*

Momentum is a vector (directional) quantity  $\therefore$  in *bound stationary quantum states* the expectation value satisfies  $\langle p \rangle = 0$ , reflecting the absence of any preferred direction of motion (momenta cancel each other in all directions).

Here,  $\langle p \rangle$  denotes the *expectation value* (the average result of repeated measurements) of the momentum  $p$ .

## ► Reasoning

- $\therefore$  momentum spread necessarily makes  $\langle p^2 \rangle > 0$  even in , kinetic energy cannot vanish.
- $\therefore$  *uncertainty* implies nonzero average *kinetic energy*.

## ► Takeaway

- Momentum spread guarantees ground-state kinetic energy.

# Why Temperature Ignores Zero-Point Energy

## ▶ Question

- ▶ Why can temperature reach zero although zero-point energy remains?

## ▶ Evidence

- ▶ Every quantum system retains *zero-point* energy in its ground state.
- ▶ Heating changes populations of excited states above the ground state.

## ▶ Reasoning

- ▶  $\therefore$  temperature measures thermal excitations, ground-state energy contributes nothing to temperature.
- ▶  $\therefore$  *absolute zero* means no thermal excitation, not zero total energy.

## ▶ Takeaway

- ▶ Temperature measures additional thermal energy above the ground state, not the unavoidable ground-state energy itself.



# Zero-Point Motion and Temperature

## ▶ Question

- ▶ How can a *single atom* have kinetic energy without having a *defined temperature*?

## ▶ Evidence

- ▶ Zero-point energy exists in a single quantum ground state.
- ▶ Temperature reflects thermal excitation above the ground state and describes the *distribution of energy* in an *equilibrium system*.

## ▶ Reasoning

- ▶ *uncertainty principle* (quantum confinement)  
⇒ kinetic energy in the ground state (zero-point energy)
- ▶ *thermal energy* ⇒ excited-state populations
- ▶ ∴ *zero-point* kinetic energy is not the atom's *temperature*.

## ▶ Takeaway

- ▶ Zero-point motion is a quantum property of a confined particle;  
*temperature* emerges only from thermal populations in equilibrium.

# Why Absolute Zero Cannot Be Reached

- ▶ **Question**

- ▶ Why can no real cooling process reach absolute zero?

- ▶ **Evidence**

- ▶ Cooling becomes progressively less effective near *absolute zero*.
  - ▶ Each cooling step removes a smaller amount of thermal energy.

- ▶ **Reasoning**

- ▶  $\therefore$  *finite* steps never eliminate the remaining thermal excitation, cooling continually slows.
  - ▶  $\therefore$  the third law of thermodynamics forbids reaching  $T = 0$  in finite processes.

- ▶ **Takeaway**

- ▶ Absolute zero is approached, never reached.

## Outline of Part II: Gas Laws

Temperature and the Gas Variables

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# Boyle's Law

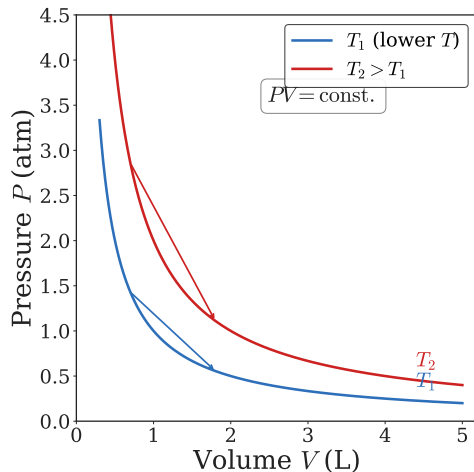
## Boyle's Law (constant $T$ , $n$ )

At constant temperature and amount, pressure and volume are **inversely proportional**:

$$V \propto \frac{1}{P} \Rightarrow PV = \text{const.}$$

$$P_1 V_1 = P_2 V_2$$

Squeezing a sealed balloon: volume  $\downarrow \Rightarrow$  pressure  $\uparrow$ .



## Check Your Understanding

A gas occupies 4.0 L at 1.0 atm.

The pressure is increased to 2.0 atm at constant temperature.

What is the new volume?

Come to lecture to see the answer.

# Deriving Boyle's Law from the Kinetic Molecular Theory

## ▶ Question

- ▶ Why does decreasing *volume* increase gas pressure?

## ▶ Evidence

- ▶ At constant temperature, molecular speed is unchanged, so elastic collisions transfer the same momentum.
- ▶ Smaller volume shortens time between successive wall collisions.

## ▶ Reasoning

- ▶  $\therefore$  momentum per collision is unchanged, only *collision rate* increases.
- ▶  $\therefore$  more momentum reaches the wall each second, increasing force.
- ▶  $\therefore$  greater force on the same area raises *pressure*.

## ▶ Takeaway

- ▶ Pressure rises because collisions occur more frequently.

# Gay-Lussac's Law

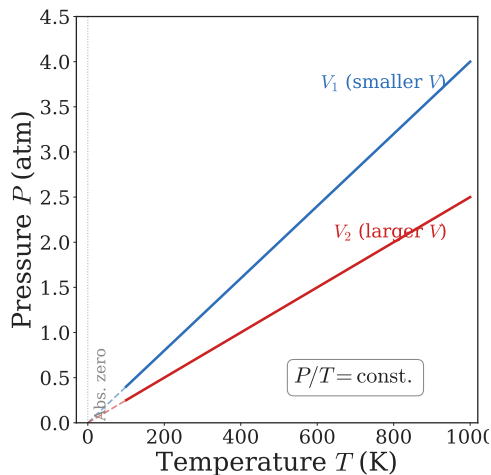
## Gay-Lussac's Law (constant $V, n$ )

At constant volume and amount, pressure is **directly proportional** to temperature (in K):

$$P \propto T \Rightarrow \frac{P}{T} = \text{const.}$$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

The pressure inside a sealed aerosol can increases if it is left in a hot car.



# Charles's Law

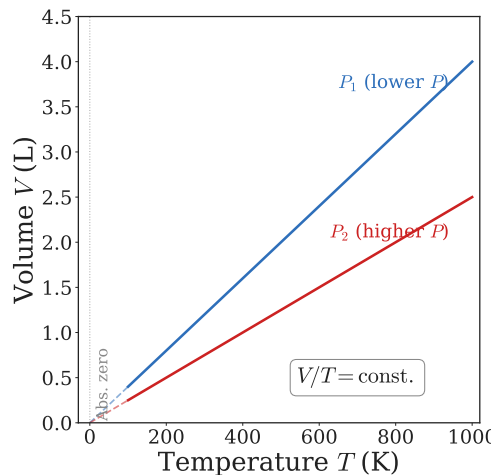
## Charles's Law (constant $P$ , $n$ )

At constant pressure and amount, volume is **directly proportional** to temperature (in K):

$$V \propto T \Rightarrow \frac{V}{T} = \text{const.}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

A balloon placed in a freezer shrinks; in a hot car, it expands.





## Check Your Understanding

A balloon holds 2.0 L at 300 K.

It is cooled to 150 K at constant pressure.

What is the new volume?

Come to lecture to see the answer.

# Why Does Heating a Gas Increase Either Pressure or Volume?

## ▶ Question

- ▶ Why does the same temperature increase produce different macroscopic changes?

## ▶ Evidence

- ▶ Heating increases molecules' *average kinetic energy*.
- ▶ Fixed-volume walls cannot move (sealed aerosol can).
- ▶ Constant-pressure walls can move (helium balloon).

## ▶ Reasoning

- ▶ Faster collisions with fixed walls increase force  $\therefore$  *pressure* rises.
- ▶ Movable walls expand until collision force matches external pressure  $\therefore$  *volume* rises instead.

## ▶ Takeaway

- ▶ Constraints determine whether heating changes pressure or volume.

# Avogadro's Law

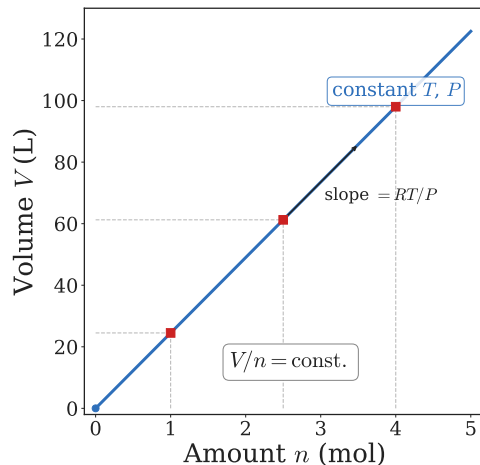
## Avogadro's Law (constant $T, P$ )

At constant temperature and pressure, volume is **directly proportional** to the amount of gas:

$$V \propto n \quad \Rightarrow \quad \frac{V}{n} = \text{const.}$$

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

A punctured tire:  
gas escapes (moles ↓) so volume ↓.



## Why Does *Gas Identity* Not Affect Equal Gas Volumes?

### ▶ Question

- ▶ Why do equal gas volumes contain equal numbers of molecules despite different chemical identities?

### ▶ Evidence

- ▶ At the same *temperature* and pressure, different gases exert identical pressure behavior.
- ▶ Gas molecules *themselves* occupy negligible volume and interact only during collisions.

### ▶ Reasoning

- ▶ Pressure depends on collision frequency and kinetic energy, not molecular identity.
- ▶  $\therefore$  equal  $T$  and  $P$  require equal particle density regardless of gas type.

### ▶ Takeaway

- ▶ *Particle number*, not identity, determines gas volume at fixed  $T, P$ .

## Summary: The Simple Gas Laws

| Name         | Relationship    | Held Const. | Example              |
|--------------|-----------------|-------------|----------------------|
| Boyle's      | $V \propto 1/P$ | $T, n$      | Squeezing a balloon  |
| Charles's    | $V \propto T$   | $P, n$      | Balloon in freezer   |
| Gay-Lussac's | $P \propto T$   | $V, n$      | Heating a sealed can |
| Avogadro's   | $V \propto n$   | $T, P$      | Punctured tire       |

Remember: Compressibility depends on the State of the Fluid

- ▶ Gases are **compressible**: volume responds to pressure changes.
- ▶ Liquids are **incompressible**: volume does not change with pressure.

## Check Your Understanding

A gas sample occupies 8.0 L and contains 2.0 mol at constant  $T$  and  $P$ .

If gas is added until the amount reaches 3.0 mol, what is the new volume?

Come to lecture to see the answer.

## Outline of Part II: Gas Laws

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# Combining the Gas Laws: Ideal-Gas Law

Combining three simple gas laws:

$$V \propto \frac{1}{P}, \quad V \propto T, \quad V \propto n \quad \Longrightarrow \quad V \propto \frac{nT}{P}$$

The value of the proportionality constant  $R$  (the **gas constant**) is found through experiment:

## The Ideal-Gas Law

$$PV = nRT$$

$$R = 0.08206 \text{ L}\cdot\text{atm}\cdot\text{mol}^{-1}\cdot\text{K}^{-1} \quad (\text{use when } P \text{ in atm, } V \text{ in L})$$

$$R = 8.314 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1} \quad (\text{SI; when } P \text{ in Pa, } V \text{ in m}^3)$$



# Deriving the Ideal Gas Equation from Kinetic Molecular Theory

## Question

- How does molecular motion produce the macroscopic relation  $PV = nRT$ ?

## Evidence

- Elastic wall collisions transfer *momentum* to the container.
- Summing all collisions gives  $P = \frac{1}{3} \frac{N}{V} m \overline{u^2}$ .

## Reasoning

- Pressure increases with molecular number density ( $\frac{N}{V}$ ) and average kinetic energy ( $2 \overline{\text{KE}} = m \overline{u^2}$ ).
- $\overline{\text{KE}} = \frac{3}{2} k_B T$  links molecular motion to *temperature*.
- $\therefore$  Replacing  $\overline{u^2}$  gives  $PV = Nk_B T = nRT$ .

## Takeaway

- Microscopic* motion  $\Rightarrow$  macroscopic gas law.

- Consider one molecule of mass  $m$  with velocity  $(u_x, u_y, u_z)$ .
- An elastic collision reverses  $u_x$ , so  $\Delta p_x = 2mu_x$ .
- The molecule strikes the same wall every  $\Delta t = 2L/u_x$ .
- Average force:  $F = \Delta p_x / \Delta t = mu_x^2 / L$ .
- Pressure from one molecule:

$$P = \frac{F}{A} = \frac{mu_x^2}{LA} = \frac{mu_x^2}{V}.$$

- For  $N$  molecules:

$$P = \frac{m}{V} \sum_i u_{x,i}^2.$$

- Random molecular motion gives

$$\overline{u_x^2} = \overline{u_y^2} = \overline{u_z^2} = \frac{1}{3} \overline{u^2}.$$

$$\therefore P = \frac{Nm \overline{u_x^2}}{V} = \frac{1}{3} \frac{N}{V} m \overline{u^2}.$$

# The Physical Meaning of $PV$

## ▶ Question

- ▶ What is  $PV$ ? What does it actually signify?

## ▶ Evidence

- ▶ Molecular collisions create the pressure on the container walls.
- ▶ Moving a wall outward by  $\Delta x$  creates a space of volume  $V = A\Delta x$ .

## ▶ Reasoning

- ▶ The gas pushes with force  $F = PA$ ; expanding the wall against pressure requires mechanical work  $W = F\Delta x = PV$ .
- ▶  $\therefore$  Kinetic molecular theory predicts  $PV = \frac{2}{3}N\overline{KE} = Nk_B T = nRT$ .

## ▶ Takeaway

- ▶  $PV$  is an energy scale:
  - ▶ Mechanically, it is the work needed to create the gas volume against pressure.
  - ▶ Microscopically, it is proportional to the total translational kinetic energy of the molecules.

# Ideal Behavior and Conditions

- ▶ The ideal-gas law assumes:
  1. Gas molecules occupy negligible volume compared with the container.
  2. Intermolecular forces are negligible except during brief collisions.
- ▶ Gases behave most **ideally** at high temperature and low pressure.

Note: I.G. Law is valid at STP (0 °C and 1 atm)

$$\text{Ideality} \propto T$$

Higher  $T$ : kinetic energy dominates over intermolecular attractions

$$\text{Ideality} \propto \frac{1}{P}$$

Lower  $P$ : far apart, rarely interact

1. The ideal-gas law assumes point particles with no intermolecular forces.
2. Low pressure increases the average separation between molecules, making interactions rare and weak.
3. High temperature increases molecular kinetic energy, making any remaining interactions relatively unimportant.
4. Under these conditions, molecules move almost independently and collide elastically.
5.  $\therefore$  the ideal-gas law provides an excellent approximation.

# When Is a Gas Effectively *Ideal*?

## Question

- ▶ Why do gases behave ideally only at high  $T$  and low  $P$ ?

## Evidence

- ▶ Compare  $k_B T$  with the energy of intermolecular attractions
- ▶ Compare volume of each molecule  $b$  with volume accessible to each molecule  $V/N$ .

## Reasoning

- ▶ if  $T$  is high, then  $k_B T \gg$  intermolecular-attraction energy, and *intermolecular attractions* are negligible.
- ▶ if  $P$  is low, then  $V/N \gg b$ , and molecular size scarcely reduces accessible volume.
- ▶  $\therefore$  neither attraction nor molecular volume significantly alters ideal-gas predictions at high  $T$  and low  $P$ .

## Takeaway

- ▶ *Ideality* depends not only on pressure and temperature, but also on the gas' intermolecular-attraction strength and molecular size.

*Energy of intermolecular attractions* is equal to the depth of the intermolecular potential well.

Remember:  $k_B T$  is the *typical thermal energy* available to one molecule due to random thermal motion.

# Standard Temperature and Pressure (STP)

## Standard Temperature and Pressure (STP)

- ▶  $T = 0\text{ }^{\circ}\text{C} = 273.15\text{ K}$
- ▶  $P = 1\text{ atm} = 101\,325\text{ Pa}$

## Standard Molar Volume

One mole of any *ideal* gas at STP occupies:

$$V = \frac{nRT}{P} = \frac{(1\text{ mol})(0.08206\text{ L}\cdot\text{atm}\cdot\text{mol}^{-1}\cdot\text{K}^{-1})(273.15\text{ K})}{1\text{ atm}} = \boxed{22.41\text{ L}}$$

$22.41\text{ L mol}^{-1}$  is the **standard molar volume** — a useful shortcut for STP problems.

## Relating Two States: The Combined Gas Law

When the amount of gas is constant ( $n$  fixed),  $PV/T$  is constant:

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

Special cases (one variable held fixed):

- ▶  $T$  constant:  $P_1 V_1 = P_2 V_2$  (Boyle's Law)
- ▶  $P$  constant:  $V_1/T_1 = V_2/T_2$  (Charles's Law)
- ▶  $V$  constant:  $P_1/T_1 = P_2/T_2$  (Gay-Lussac's Law)

## Example: Relating Two States

State 1:  $P_1 = 1.00 \text{ atm}$ ,  $T_1 = 25.00^\circ\text{C}$ ,  $V_1 = 1.00 \text{ L}$ .

Cool to  $T_2 = -5.00^\circ\text{C}$  at constant pressure. Find  $V_2$ .

$$T_1 = 298.15 \text{ K}, \quad T_2 = 268.15 \text{ K}$$

$$P_1 = P_2 \text{ (constant)}$$

---


$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$V_2 = V_1 \cdot \frac{T_2}{T_1} \cdot \underbrace{\frac{P_1}{P_2}}_{=1}$$

$$V_2 = 1.00 \text{ L} \cdot \frac{268.15 \text{ K}}{298.15 \text{ K}}$$

$V_2 = 0.899 \text{ L}$

- Always convert  $T$  to Kelvin first!
- Cooling  $\Rightarrow$  volume decreases (Charles's Law)

## Example: Using the Ideal-Gas Law

### Scuba Tank

A scuba tank contains 13.2 L of air at 153 atm.

What volume of air at 1.00 atm and 25.0 °C does this represent?

(Use combined gas law,  $n$  constant.)

$$P_1 = 153 \text{ atm}, \quad V_1 = 13.2 \text{ L}, \quad T_1 = 298 \text{ K}$$

$$P_2 = 1.00 \text{ atm}, \quad V_2 = ?, \quad T_2 = 298 \text{ K}$$

---

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$V_2 = \frac{P_1 V_1 T_2}{T_1 P_2} = \frac{(153 \text{ atm})(13.2 \text{ L})(298 \text{ K})}{(298 \text{ K})(1.00 \text{ atm})}$$

$V_2 = 2020 \text{ L}$

- ▶  $T_1 = T_2$ , so the temperature ratio cancels.
- ▶ 2020 L at atmospheric pressure — a huge volume for diving!



## Density and Molar Mass from the Ideal-Gas Law

### Density of an Ideal Gas

From  $PV = nRT$  and  $n = m/\mathcal{M}$ :

$$\rho = \frac{m}{V} = \frac{P\mathcal{M}}{RT}$$

where  $\mathcal{M}$  is molar mass.

Higher  $P$  or lower  $T \Rightarrow$  denser gas.

### Molar Mass of an Unknown Gas

Rearranging  $PV = \frac{m}{\mathcal{M}}RT$ :

$$\mathcal{M} = \frac{mRT}{PV}$$

Measure  $m$ ,  $P$ ,  $V$ ,  $T$  to identify an unknown gas.

- These formulas let you use a gas as if it were a solid sample: measure its mass and gas properties, then find its molar mass.

## Check Your Understanding

At the same temperature and pressure, which gas is **densest**?

(A) He    (B) H<sub>2</sub>    (C) N<sub>2</sub>    (D) CO<sub>2</sub>

Come to lecture to see the answer.

# Density and Molar Mass from the Ideal-Gas Law

## Question

- ▶ Why do ideal gases have different densities under the same pressure and temperature?

## Evidence

- ▶ At fixed  $P$ ,  $T$ , equal gas volumes contain equal numbers of molecules.
- ▶ Different gases still have different masses in those equal volumes.

## Reasoning

- ▶  $\therefore$  *number density* is fixed, molecule count cannot explain density differences.
- ▶  $\therefore$  Only *mass per molecule* changes the mass within the same volume.
- ▶  $\therefore \rho = PM/RT$ : density is proportional to *molar mass*.

## Takeaway

- ▶ Density reveals molar mass at fixed pressure and temperature.

## Gas Densimetry for Determining Molar Mass — (19th century)

### Question

- ▶ How can we experimentally determine a gas's molar mass?

### Evidence

- ▶ Measure a gas's mass in a container of known volume.
- ▶ Record pressure and temperature to obtain *density*.

### Reasoning

- ▶  $\because \rho = m/V$ , the experiment determines density.
- ▶  $\therefore \mathcal{M} = \rho RT/P$  gives the *molar mass*.
- ▶  $\therefore$  Molecular formulas were tested by comparing calculated and predicted molar masses.

### Takeaway

- ▶ Gas densimetry linked measurements to molecular formulas.

## Part III

# Gas Mixtures

# Outline of Part III: Gas Mixtures

Dalton's Law and Partial Pressures

Gas Stoichiometry

# Outline of Part III: Gas Mixtures

Dalton's Law and Partial Pressures

Gas Stoichiometry

# Mixtures of Gases: Dalton's Law

## Dalton's Law of Partial Pressures

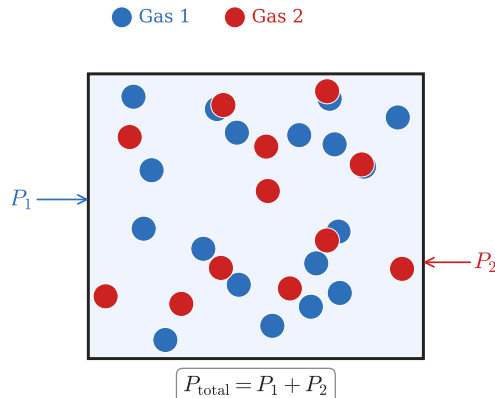
Each gas in a mixture exerts pressure **independently** of the others.

The **total pressure** equals the sum of all **partial pressures**:

$$P_{\text{total}} = \sum_i P_i = P_1 + P_2 + P_3 + \cdots$$

## Dry Air at 760 mmHg

$$P_{\text{N}_2} \approx 593, P_{\text{O}_2} \approx 160, P_{\text{Ar}} \approx 7 \text{ mmHg}$$
$$P_{\text{total}} \approx 760 \text{ mmHg} \checkmark$$





# Why Do Mixed Gases Exert Independent Pressures?

## Question

- ▶ Why does adding one gas not change another gas's pressure contribution?

## Evidence

- ▶ Gas pressure arises from molecule–wall collisions.
- ▶ *KMT postulate 4*: gas particles exert negligible intermolecular forces.
- ▶ Each gas keeps its own collision frequency and momentum transfer.

## Reasoning

- ▶  $\therefore$  particles do not interact, other gases cannot alter a gas's collisions.
- ▶  $\therefore$  each gas contributes an *independent partial pressure*.
- ▶ Strong intermolecular forces would change collision behavior, so independence would fail.

## Takeaway

- ▶ Dalton's law follows directly from negligible intermolecular forces.

# When Does Dalton's Law Break Down?

## Question

- ▶ Why do some gas mixtures not obey Dalton's law?

## Evidence

- ▶ Real gases develop significant *intermolecular forces* at high pressure or low temperature. *Why in these regimes?*
- ▶ These forces change molecular trajectories and momentum transfer.

## Reasoning

- ▶  $\therefore$  molecules interact, one gas alters another gas's collision behavior.
- ▶ Independent *partial pressures* no longer exist.
- ▶  $\therefore$  Dalton's law fails when KMT postulate 4 is violated.

## Takeaway

- ▶ *Gas interactions* determine Dalton's law's limits.

# How Do Attractions Change the Measured Pressure?

## Question

- ▶ How do attractive forces affect the total pressure? Is it greater or lower than Dalton's law predicts?

## Evidence

- ▶ Attractive forces pull approaching molecules back from the wall.
- ▶ Wall collisions become less frequent or less forceful.

## Reasoning

- ▶  $\therefore$  attractions pull molecules back from the wall, they transfer less momentum during collisions.
- ▶ Each gas therefore contributes less pressure than it would in an ideal mixture.
- ▶  $\therefore$  the measured pressure is lower than Dalton's law predicts.

## Takeaway

- ▶ *Attractive forces* lower pressure below Dalton's law.

# Mole Fraction and Partial Pressure

## Definition: Mole Fraction ( $\chi_i$ )

The mole fraction of component  $i$  is the fraction of total moles it contributes:

$$\chi_i = \frac{n_i}{n_{\text{total}}}, \quad \sum_i \chi_i = 1$$

## Partial Pressure from Mole Fraction

$$P_i = \chi_i \times P_{\text{total}}$$

$$\text{e.g. } \chi_3 = \frac{1}{6} = 0.167 \Rightarrow P_3 = 0.167 \times P_{\text{total}}$$

## Why does $P_i = \chi_i P_{\text{total}}$ Work?

Pressure arises from particle-wall momentum transfer.



Temperature fixes average molecular kinetic energy.



Equal temperatures make each species' momentum statistics equal.



Difference in species lies in number of molecules\*.



More molecules cause proportionally more wall collisions.



Hence,  $P_i \propto N_i$ .



Also,  $P_{\text{total}} \propto N_{\text{total}}$ .



Compare component with the whole mixture.



$$P_i = \chi_i P_{\text{total}}, \quad \chi_i = \frac{N_i}{N_{\text{total}}}$$

\* Since we are assuming these are ideal gases, we assume:

- Negligible interatomic interactions
- Negligible size of each molecule

With these assumptions, what distinguishes component gas A from component gas B is only the *number of their molecules* in the sample.

## Example: Partial Pressures

A gas mixture has  $P_{\text{total}} = 2.0 \text{ atm}$ . The mole fraction of gas 3 is  $\chi_3 = \frac{1}{6}$ . Find  $P_3$ .

$$P_3 = \chi_3 \times P_{\text{total}}$$

$$P_3 = \left(\frac{1}{6}\right)(2.0 \text{ atm})$$

$P_3 = 0.33 \text{ atm}$

Gas 3 is  $\frac{1}{6} \approx 16.7\%$  of the mixture by moles, so it contributes 16.7% of the total pressure.

## Check Your Understanding

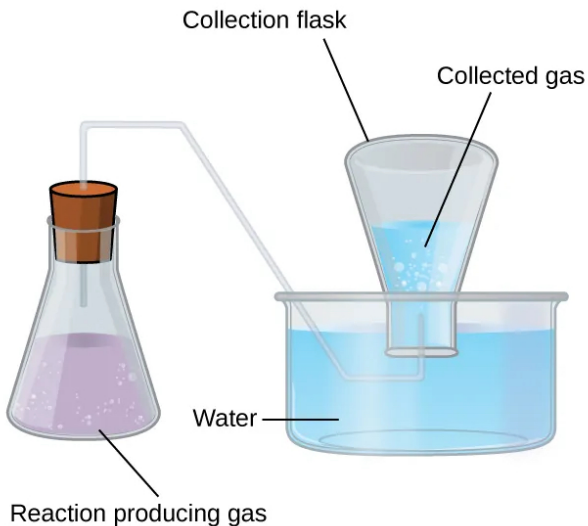
A container holds three gases at  $P_{\text{total}} = 4.0 \text{ atm}$ :

$$\chi_A = 0.50, \quad \chi_B = 0.30, \quad \chi_C = 0.20$$

Find  $P_A$ ,  $P_B$ , and  $P_C$ .

Come to lecture to see the answer.

## Measuring a Gas without it Mixing with Air: Collecting Gas over Water





## Measuring a Gas without it Mixing with Air: Collecting Gas over Water

**Purpose:** collecting gas over water allows us to capture and measure a gas as it is produced without letting it mix with the surrounding air.

1. A chemical reaction produces a gas.
2. The gas is bubbled into an inverted container filled with water.
3. As the gas enters, it displaces (pushes out) an equal volume of water.
4. The gas becomes trapped at the top of the container, allowing its volume to be measured.

The water is not there to react with the gas. It serves as a convenient, movable barrier that initially fills the container so there is no air inside.

When gas is collected by water displacement, the collected gas is saturated with water vapor:

$$P_{\text{total}} = P_{\text{gas}} + P_{\text{H}_2\text{O}}$$

Pressure of the **dry** gas:

$$P_{\text{gas}} = P_{\text{total}} - P_{\text{H}_2\text{O}}$$

- $P_{\text{H}_2\text{O}}$  depends on temperature (found in tables).
- At 25 °C:  $P_{\text{H}_2\text{O}} = 23.8 \text{ mmHg}$ .
- At 100 °C:  $P_{\text{H}_2\text{O}} = 760 \text{ mmHg}$
- a liquid boils when its vapor pressure equals the external pressure.  
(at 1 atm, water boils at 100 °C)

## How Does a Sealed Liquid Produce *Vapor Pressure*?

### ▶ Question

- ▶ Why does the pressure above a newly sealed liquid rise, then become constant?

### ▶ Evidence

- ▶ Initially, few vapor molecules occupy the space above the liquid.
- ▶ Escaping molecules increase vapor pressure and the return rate.

### ▶ Reasoning

- ▶  $\therefore$  Pressure stops increasing, vapor no longer accumulates.
- ▶  $\therefore$  Escape and return rates are equal: *dynamic equilibrium*.
- ▶  $\therefore$  The constant pressure is the liquid's *vapor pressure*.

### ▶ Takeaway

- ▶ Vapor pressure is the equilibrium pressure above a liquid.

## Why Doesn't Adding Air Change Water's *Vapor Pressure*?

### ▶ Question

- ▶ Why does adding air above liquid water not change its vapor pressure?

### ▶ Evidence

- ▶ Liquid molecules continuously evaporate and condense.
- ▶ Other gases do not undergo the liquid-vapor phase change.

### ▶ Reasoning

- ▶ Only the liquid's molecules contribute to both the escape and return fluxes.
- ▶  $\therefore$  Only those molecules establish the *dynamic equilibrium* that fixes the equilibrium pressure.
- ▶  $\therefore$  Other gases change the total pressure, but not the liquid's *vapor pressure*.

### ▶ Takeaway

- ▶ A liquid's vapor pressure *only* depends on temperature.

## Why Does a Liquid Boil When Its *Vapor Pressure* Equals the External Pressure?

### ▶ Question

- ▶ Why does a liquid begin boiling only when its *vapor pressure* equals the external pressure?

### ▶ Evidence

- ▶ Tiny vapor bubbles constantly form inside the liquid.
- ▶ When external pressure exceeds the vapor pressure (which is the bubbles' internal pressure), the bubbles collapse.

### ▶ Reasoning

- ▶  $\therefore$  Heating increases the liquid's *vapor pressure*, bubble pressure increases.
- ▶  $\therefore$  When vapor pressure equals the external pressure, bubbles become stable and expand.
- ▶  $\therefore$  The liquid boils throughout, not only at its surface.

### ▶ Takeaway

- ▶ Boiling begins when vapor bubbles become stable inside the liquid.

# Outline of Part III: Gas Mixtures

Dalton's Law and Partial Pressures

Gas Stoichiometry

# Gas Stoichiometry

The ideal-gas law connects moles to measurable gas quantities:

$$n = \frac{PV}{RT}$$

This lets us use a gas volume the way we use the mass of a solid in stoichiometry.

## General Approach

1. Convert known gas volume  $\rightarrow$  moles using  $n = PV/RT$  (or use STP shortcut).
2. Apply stoichiometric mole ratios from the balanced equation.
3. Convert product moles  $\rightarrow$  volume using  $V = nRT/P$ .

(Shortcut)  $\therefore$  Known Molar Volume At STP:  $n = V/(22.41 \text{ L mol}^{-1})$ .

## Check Your Understanding

For the reaction:  $\text{N}_2(\text{g}) + 3 \text{H}_2(\text{g}) \rightarrow 2 \text{NH}_3(\text{g})$

All volumes measured at the same  $T$  and  $P$ .

How many liters of  $\text{NH}_3$  are produced from 5.0 L of  $\text{N}_2$ ?

Come to lecture to see the answer.

## Part IV

# Kinetic Molecular Theory



## Outline of Part IV: Kinetic Molecular Theory

Effusion and Diffusion

Kinetic Molecular Theory

# Outline of Part IV: Kinetic Molecular Theory

Effusion and Diffusion

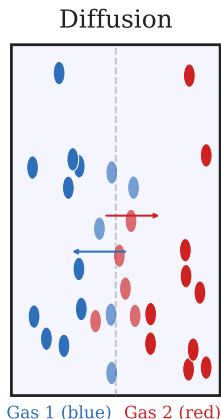
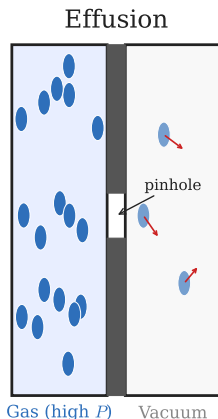
Kinetic Molecular Theory

# Effusion and Diffusion

## Effusion

The escape of gas molecules through a **tiny pinhole** into a vacuum or lower-pressure region.

A helium balloon slowly deflates as He atoms effuse through microscopic pores in the latex.



## Diffusion

The spreading of gas molecules through space in response to a **concentration gradient** (high  $\rightarrow$  low concentration).

The scent of perfume gradually spreads across a room by diffusion.

# Why Does Graham's Law Predict Effusion Better Than Diffusion?

## Question

- ▶ Why does Graham's Law describe *effusion* accurately but only approximate *diffusion*?

## Evidence

- ▶ Effusion passes through a tiny hole into a vacuum.
- ▶ Diffusion occurs through a gas along a concentration gradient.
- ▶ Diffusing molecules repeatedly collide, changing their paths.

## Reasoning

- ▶  $\therefore$  effusion involves essentially no intermolecular collisions after the hole, escape depends mainly on molecular speed.
- ▶  $\therefore$  diffusion also depends on *mean free path*, collisions reduce the influence of speed.
- ▶  $\therefore$  Graham's Law predicts effusion directly but diffusion only approximately.

## Takeaway

- ▶ *Intermolecular collisions* limit Graham's Law during diffusion.

## Which Factors Determine the Rate of Effusion?

### ▶ Question

- ▶ Which factors determine how fast helium effuses through a tiny pinhole?

### ▶ Evidence

- ▶ Higher *temperature* makes molecules strike the hole more often and faster.
- ▶ Larger pressure difference sends more molecules toward the hole.
- ▶ A larger pinhole allows more molecules to escape simultaneously.

### ▶ Reasoning

- ▶  $\therefore$  effusion depends on impacts at the opening, only factors changing those impacts affect the rate.
- ▶  $\therefore$  temperature, pressure difference, and pinhole size matter.

### ▶ Takeaway

- ▶ Effusion rate is set by molecular impacts at the hole.

# Why Is Gas Diffusion Fast Locally but Slow Across a Room?

## ▶ Question

- ▶ Why do gas molecules spread rapidly over short distances but slowly across a room?

## ▶ Evidence

- ▶ Gas molecules move at high *RMS speed*.
- ▶ They frequently collide, giving a short *mean free path*.

## ▶ Reasoning

- ▶  $\therefore$  Collisions constantly randomize motion, fast molecules rarely travel far in one direction.
- ▶  $\therefore$  Diffusion depends on both *mean free path* and RMS speed, not speed alone.
- ▶  $\therefore$  Rapid microscopic motion produces slow macroscopic spreading.

## ▶ Takeaway

- ▶ Collisions limit diffusion despite rapid molecular motion.

# Graham's Law of Effusion

## Graham's Law

The rate of effusion of a gas is **inversely proportional** to the square root of its molar mass:

$$\text{rate} \propto \frac{1}{\sqrt{\mathcal{M}}}$$

For two gases A and B:

$$\frac{\text{rate}_A}{\text{rate}_B} = \sqrt{\frac{\mathcal{M}_B}{\mathcal{M}_A}}$$

**Lighter gases move faster**  $\Rightarrow$  they effuse and diffuse more rapidly.  
This is because lighter molecules must travel faster to have the same kinetic energy.

## Example: Graham's Law

### H<sub>2</sub> vs. O<sub>2</sub> Effusion Rate

How much faster does H<sub>2</sub> effuse than O<sub>2</sub>?

( $\mathcal{M}_{\text{H}_2} = 2.016 \text{ g mol}^{-1}$ ,  $\mathcal{M}_{\text{O}_2} = 32.00 \text{ g mol}^{-1}$ )

$$\begin{aligned}\frac{\text{rate}_{\text{H}_2}}{\text{rate}_{\text{O}_2}} &= \sqrt{\frac{\mathcal{M}_{\text{O}_2}}{\mathcal{M}_{\text{H}_2}}} = \sqrt{\frac{32.00 \text{ g mol}^{-1}}{2.016 \text{ g mol}^{-1}}} \\ &= \sqrt{15.87}\end{aligned}$$

$$\frac{\text{rate}_{\text{H}_2}}{\text{rate}_{\text{O}_2}} \approx 3.98 \approx 4$$

H<sub>2</sub> effuses  $\approx 4$  times faster than O<sub>2</sub>.

Intuition:  $\sqrt{32/2} = \sqrt{16} = 4$ .



## Check Your Understanding

Which effuses faster: He ( $\mathcal{M} = 4.00 \text{ g mol}^{-1}$ ) or N<sub>2</sub> ( $\mathcal{M} = 28.0 \text{ g mol}^{-1}$ )?

By approximately what factor?

Come to lecture to see the answer.

# Why Do Lighter Gases Escape Faster?

## Question

- ▶ Why do lighter gases pass through a tiny hole more rapidly at the same temperature?

## Evidence

- ▶ All gases at equilibrium have the same average *kinetic energy*:  $\overline{KE} = \frac{3}{2}k_B T$ .
- ▶ Molecular kinetic energy is  $\frac{1}{2}mu^2$ .
- ▶ Molecules reach the hole by random motion; faster molecules encounter it more often.

## Reasoning

- ▶  $\therefore \frac{1}{2}mu^2$  is equal for all gases, lighter molecules must have larger  $u$ .
- ▶  $\therefore$  encounter rate with the hole is proportional to *speed*:  $\text{rate} \propto u$ .
- ▶  $\therefore \text{rate} \propto 1/\sqrt{M}$ , giving *Graham's law*.

## Takeaway

- ▶ Equal kinetic energy makes lighter gases effuse faster.

# When Does Graham's Law Apply? Does Pressure Matter?

## Question

- ▶ Does changing pressure prevent Graham's law from predicting effusion rates?

## Evidence

- ▶ Effusion rate equals *number density*  $\times$  molecular speed at the hole.
- ▶ Equal temperature gives the same average kinetic energy, so  $v \propto 1/\sqrt{M}$ .
- ▶ Different pressures produce different number densities.

## Reasoning

- ▶  $\therefore$  rate depends on both density and speed, mass alone cannot determine effusion.
- ▶  $\therefore$  Graham's law requires *equal pressure* so molecular density is the same.
- ▶  $\therefore$  only *molar mass* remains to control the rate ratio.

## Takeaway

- ▶ Equal pressure is essential for Graham's law.

# Why Does Effusion Rate Scale with the Square Root of Molar Mass?

- ▶ Question
  - ▶ Why do lighter gases effuse faster with rate  $\propto 1/\sqrt{\mathcal{M}}$ ?
- ▶ Evidence
  - ▶ At equilibrium, all gases have the same average *kinetic energy*.
  - ▶ Kinetic energy satisfies  $\frac{1}{2}mu^2 = \frac{3}{2}k_B T$ .
- ▶ Reasoning
  - ▶  $\therefore$  kinetic energy depends on  $u^2$ , equal energy requires  $v \propto 1/\sqrt{m}$ .
  - ▶ Molecules reach the hole by motion, not mass alone.
  - ▶  $\therefore$  *effusion rate*  $\propto v \propto 1/\sqrt{\mathcal{M}}$ .
- ▶ Takeaway
  - ▶ The square root comes from kinetic energy's quadratic speed dependence.

# Why Balloons Slowly Deflate

Gas Permeation: Solubility and Diffusivity determine Rate of Diffusion

Left : Helium  
 Right : Argon

Time: 0 hr



Time: 12 hr



Why does the He balloon deflate more than the Ar balloon?

$$P = S \times D$$

$P$ : permeability  
 $S$ : solubility  
 $D$ : diffusivity

1. The balloon is made of long, flexible polymer chains, while the gas inside consists of molecules in constant thermal motion.
2. The polymer chains continually fluctuate, creating temporary intermolecular spaces.
3. Gas molecules constantly collide with the polymer surface.
4. Occasionally, a gas molecule enters a temporary intermolecular space and becomes *dissolved in the polymer*.
5. The polymer chains continually move, opening neighboring spaces into which the gas molecule can move, allowing it to *diffuse through the polymer*.
6. The gas concentration is higher inside than outside  $\Rightarrow$  *more molecules diffuse outward than inward* through the polymer.
7. Gas molecules emerge from the outer surface  $\Rightarrow$  the balloon *gradually loses gas*.
8. As gas escapes, the internal gas pressure decreases.
9. Less internal pressure stretches the polymer  $\Rightarrow$  the balloon contracts and *gradually deflates*.

# Why Balloons Slowly Deflate

Gas Permeation: Solubility and Diffusivity determine Rate of Diffusion

Left : Helium  
Right : Argon



Time: 0 hr

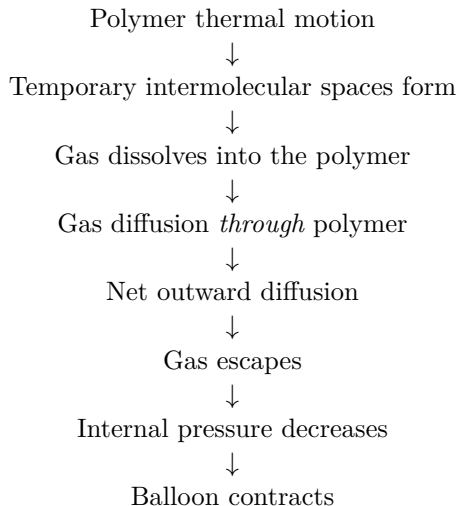


Time: 12 hr

Why does the  
He balloon  
deflate more  
than the Ar  
balloon?

$$P = S \times D$$

$P$ : permeability  
 $S$ : solubility  
 $D$ : diffusivity



# Why Does Helium Escape a Balloon Faster Than Argon?

- ▶ Question
  - ▶ Why does helium permeate a balloon faster although both gases are small?
- ▶ Evidence
  - ▶ Gas crosses solid polymer by dissolving, diffusing, then desorbing.
  - ▶ Permeability follows  $P = S \times D$ , combining *solubility* and *diffusivity*.
- ▶ Reasoning
  - ▶  $\therefore$  transport occurs through polymer, effusion assumptions and Graham's Law no longer apply.
  - ▶  $\therefore$  helium dissolves better, higher  $S$  increases  $P$  despite similar size;  $\therefore$  helium escapes faster.
- ▶ Takeaway
  - ▶ *Permeation* depends on solubility and diffusivity.

# Why Is Helium More Soluble Than Argon?

- ▶ Question
  - ▶ Why does helium dissolve into balloon polymers more readily than argon?
- ▶ Evidence
  - ▶ Helium has a compact 1s electron cloud; argon has a larger, more polarizable cloud.
  - ▶ Polymer dissolution balances cavity formation against *dispersion* stabilization.
- ▶ Reasoning
  - ▶  $\therefore$  helium needs a much smaller cavity, its energetic penalty decreases more than weaker dispersion increases it.
  - ▶  $\therefore$  helium has greater *solubility*, increasing *permeability*.
- ▶ Takeaway
  - ▶ Solubility reflects competing quantum-size and interaction effects.



# Why Does the Balloon Contract Instead of Imploding?

- ▶ Question
  - ▶ Why does a leaking helium balloon shrink smoothly instead of collapsing inward?
- ▶ Evidence
  - ▶ Escaping helium lowers the pressure difference across the polymer.
  - ▶ The polymer is *elastic*, so its shape changes under changing stress.
- ▶ Reasoning
  - ▶  $\therefore$  decreasing pressure reduces membrane tension, inward collapse is no longer driven.
  - ▶  $\therefore$  the elastic polymer relaxes to a smaller equilibrium shape instead of *imploding*.
- ▶ Takeaway
  - ▶ *Elasticity* determines how the balloon shrinks.

# Outline of Part IV: Kinetic Molecular Theory

Effusion and Diffusion

Kinetic Molecular Theory

# The Kinetic Molecular Theory of Gases

## Postulates of KMT

1. Gas particles are in **continuous random motion**, moving in straight lines between collisions.
2. The volume of gas particles is **negligibly small** compared to the distances between them.
3. Gas pressure arises from **collisions** of particles with the container walls.
4. Gas particles exert **no attractive or repulsive forces** on each other.
5. All collisions are **elastic** — no kinetic energy is lost.
6. The average kinetic energy of particles is **directly proportional to the Kelvin temperature**:  $\overline{KE} \propto T$

## How KMT Explains the Gas Laws

| Gas Law      | KMT Explanation                                                                                |
|--------------|------------------------------------------------------------------------------------------------|
| Boyle's      | $V \downarrow \Rightarrow$ more wall collisions per area $\Rightarrow P \uparrow$              |
| Charles's    | $T \uparrow \Rightarrow$ faster particles; expand $V$ to keep $P$ constant                     |
| Avogadro's   | More particles $\Rightarrow$ more collisions; expand $V$ to keep $P$ constant                  |
| Dalton's     | No intermolecular forces: each gas hits walls independently                                    |
| Gay-Lussac's | $T \uparrow \Rightarrow$ harder/more frequent collisions $\Rightarrow P \uparrow$ at const $V$ |

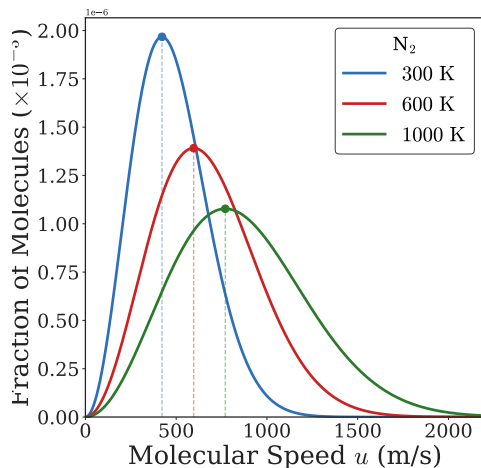
## How KMT Explains the Gas Laws

| Gas Law      | Which Quantity Changes in $P = \frac{1}{3} \left( \frac{N}{V} \right) m \overline{u^2}$ ?             |
|--------------|-------------------------------------------------------------------------------------------------------|
| Boyle's      | $V \downarrow \Rightarrow (N/V) \uparrow \Rightarrow P \uparrow$                                      |
| Charles's    | $T \uparrow \Rightarrow \overline{u^2} \uparrow$ ; $V \uparrow$ keeps $(N/V) \overline{u^2}$ constant |
| Avogadro's   | $N \uparrow \Rightarrow (N/V) \uparrow$ ; $V \uparrow$ keeps $(N/V)$ constant                         |
| Dalton's     | Each gas contributes its own $(N/V)$ independently                                                    |
| Gay-Lussac's | $T \uparrow \Rightarrow \overline{u^2} \uparrow \Rightarrow P \uparrow$ at constant $V$               |

# The Maxwell-Boltzmann Speed Distribution

Gas molecules do not all travel at the same speed. The **Maxwell-Boltzmann distribution** shows the fraction of molecules at each speed.

- ▶ **Higher  $T$ :** peak shifts to higher speeds; curve broadens and flattens.
- ▶ **Higher  $M$ :** peak shifts to *lower* speeds (heavier  $\Rightarrow$  slower at same KE).



## Check Your Understanding

Two gases at the same temperature: He and Xe.  
Which has the higher average molecular speed? Explain why.

Come to lecture to see the answer.

# Why Do Gas Molecules Have a Speed Distribution?

- ▶ Question
  - ▶ Why does a gas at thermal equilibrium contain molecules with many speeds instead of one common speed?
- ▶ Evidence
  - ▶ Collisions continually exchange energy while conserving the gas's total energy.
  - ▶ At equilibrium, quantum states are populated according to the *Boltzmann factor*,  $e^{-E/k_B T}$ .
- ▶ Reasoning
  - ▶  $\therefore$  collisions redistribute molecules among many allowed *quantum states*, equal speeds cannot remain stable.
  - ▶  $\therefore$  Boltzmann-weighted energies and all possible velocity directions produce the *Maxwell-Boltzmann* speed distribution.
- ▶ Takeaway
  - ▶ Classical gas speeds emerge from quantum equilibrium statistics.



# Why Does the Boltzmann Factor Describe Equilibrium?

- ▶ Question
  - ▶ Why are higher-energy states less probable at thermal equilibrium?
- ▶ Evidence
  - ▶ A system continually exchanges energy with a much larger thermal reservoir.
  - ▶ Lower system energies allow the reservoir more microscopic arrangements (*microstates*).
- ▶ Reasoning
  - ▶  $\therefore$  equilibrium overwhelmingly favors the most probable combined system–reservoir arrangement.
  - ▶  $\therefore$  states are weighted by the *Boltzmann factor*,  $P(E) \propto e^{-E/k_B T}$ .
  - ▶ Higher energies correspond to exponentially fewer equilibrium realizations.
- ▶ Takeaway
  - ▶ *Boltzmann weighting* reflects equilibrium probability.

# Why Does the Speed Distribution Broaden at Higher Temperature?

- ▶ Question
  - ▶ Why does increasing temperature broaden the speed distribution instead of simply shifting it?
- ▶ Evidence
  - ▶ Higher  $T$  weakens the exponential decrease in the *Boltzmann factor*,  $e^{-E/k_B T}$ .
  - ▶ Consequently, more high-energy states acquire appreciable equilibrium probabilities.
- ▶ Reasoning
  - ▶  $\therefore$  many more energies become significantly populated, molecules occupy a wider range of speeds.
  - ▶  $\therefore$  the distribution both broadens and flattens, increasing the *energy variance* rather than merely shifting.
- ▶ Takeaway
  - ▶ Higher temperature increases the spread of molecular energies.

# Why Doesn't the Maxwell–Boltzmann Distribution Peak at Zero Speed?

## ▶ Question

- ▶ Since low energy is most probable, why are molecules with zero speed *not* the most common?

## ▶ Evidence

- ▶ The Boltzmann factor favors low kinetic energies.
- ▶ The number of distinct velocity states (of speed  $v$ ) grows as  $4\pi u^2$ .

## ▶ Reasoning

- ▶ Very slow molecules have few possible velocity states despite their favorable energy.
- ▶ Very fast molecules have many states but are strongly suppressed by  $e^{-E/k_B T}$ .
- ▶  $\therefore$  the product of these effects is largest at an intermediate speed, producing the Maxwell–Boltzmann peak.

## ▶ Takeaway

- ▶ The most probable speed results from the competition between *state multiplicity* ( $u^2$ ) and *Boltzmann probability*.

# Root-Mean-Square Speed

## RMS Speed ( $u_{\text{rms}}$ )

The most commonly used average molecular speed:

$$u_{\text{rms}} = \sqrt{\frac{3RT}{\mathcal{M}}}$$

$R = 8.314 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$ ;  $T$  in K;  $\mathcal{M}$  in  $\text{kg mol}^{-1}$ ; result in  $\text{m s}^{-1}$ .

1. Gas molecules move at many different speeds.
2. Their kinetic energy depends on the square of speed:  $\text{KE} = \frac{1}{2}mu^2$
3. Therefore, the physically meaningful average is the average of  $u^2$ , not of  $u$ .
4. Taking the square root converts the averaged squared quantity into units of speed.
5. The resulting RMS speed is the speed that would give the same average kinetic energy as the actual distribution of molecular speeds.

For an ideal gas,  $u_{\text{most probable}} < u_{\text{avg}} < u_{\text{rms}}$

## Example: RMS Speed Calculation

### RMS Speed of N<sub>2</sub> at 30 °C

Calculate  $u_{\text{rms}}$  for nitrogen gas at 30 °C.

$$T = 303 \text{ K}, \quad \mathcal{M}_{\text{N}_2} = 0.02801 \text{ kg mol}^{-1}$$

$$u_{\text{rms}} = \sqrt{\frac{3RT}{\mathcal{M}}}$$

$$= \sqrt{\frac{3(8.314)(303)}{0.02801}}$$

$$u_{\text{rms}} = 519 \text{ m s}^{-1}$$

- ▶ 519 m/s  $\approx$  1160 mph!
- ▶ Very fast — explains rapid mixing of gases.
- ▶ Real diffusion is slower due to molecular collisions.

## Check Your Understanding

If the temperature of a gas doubles from 300 K to 600 K, by what factor does  $u_{\text{rms}}$  change?

Come to lecture to see the answer.

## Why Does the Number 3 Appear in $u_{\text{rms}} = \sqrt{3RT/\mathcal{M}}$ ?

- ▶ Question
  - ▶ Why does  $u_{\text{rms}}$  contain the factor 3?
- ▶ Evidence
  - ▶ Gas molecules translate independently along three perpendicular directions.
  - ▶ Each translational direction contributes equal average kinetic energy by *equipartition*.
- ▶ Reasoning
  - ▶  $\therefore$  Three equal contributions add, total translational energy is three times one-direction energy.
  - ▶  $\therefore \langle u^2 \rangle = 3RT/\mathcal{M}$ , giving  $u_{\text{rms}} = \sqrt{3RT/\mathcal{M}}$ ; fewer dimensions give smaller factors.
- ▶ Takeaway
  - ▶ The factor 3 counts translational dimensions.

# Why Does Temperature Couple to Translational Kinetic Energy?

- ▶ Question
  - ▶ Why does temperature couple to *translational* kinetic energy rather than rotational or vibrational motion?
- ▶ Evidence
  - ▶ Gas pressure arises from molecular collisions with container walls.
  - ▶ Only translational motion carries molecules to walls and transfers momentum.
- ▶ Reasoning
  - ▶  $\therefore$  only center-of-mass motion transfers momentum to walls, internal motion (rotational and vibrational) cannot produce gas pressure.
  - ▶  $\therefore$  *temperature* couples to translational kinetic energy, giving  $u_{\text{rms}}$ .
- ▶ Takeaway
  - ▶ Temperature couples to translational kinetic energy.



## Why Is $u_{\text{rms}} > \bar{u} > u_{\text{mp}}$ ?

- ▶ Question
  - ▶ Why are the three characteristic speeds different for the same Maxwell–Boltzmann distribution?
- ▶ Evidence
  - ▶ Beyond the peak, Boltzmann probability decreases gradually, leaving a long high-speed tail.
  - ▶ Only a few molecules occupy the tail, but their speeds are much larger.
- ▶ Reasoning
  - ▶  $\therefore$  rare fast molecules shift averages rightward, squaring speed emphasizes them even more.
  - ▶  $\therefore u_{\text{mp}} < \bar{u} < u_{\text{rms}}$ .
- ▶ Takeaway
  - ▶ The high-speed tail separates the characteristic speeds.

## Part V

# Non-ideal Gas Behavior

## Outline of Part V: Non-ideal Gas Behavior

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### Non-ideal Gas Behavior

## When Does the Ideal-Gas Law Break Down?

### Conditions for Non-ideal Behavior

- ▶ **High pressure:** molecules are crowded together; their finite volume is no longer negligible; repulsions become significant.
- ▶ **Low temperature:** slow-moving molecules feel intermolecular attractions more strongly.

### Two KMT Postulates That Fail

- ▶ Postulate 2 (negligible molecular volume) breaks down at **high**  $P$ .
- ▶ Postulate 4 (no intermolecular forces) breaks down at **low**  $T$ .

Most ideal: **low**  $P$  and **high**  $T$

# The van der Waals Equation: Correcting Non-ideal Behavior

Modifies the ideal-gas law with two correction terms,  $a$  and  $b$ :

$$\left( P + \left( \frac{n}{V} \right)^2 a \right) (V - nb) = nRT$$

## Constant $a$

Corrects for **intermolecular attractions**.

Larger  $a \Rightarrow$  stronger attractions.

The  $n^2a/V^2$  term adds to  $P$   
 ( $\because$  attractions reduce the effective pressure).

## Constant $b$

Corrects for **finite molecular volume**.

Larger  $b \Rightarrow$  larger molecules.

The  $nb$  term reduces  $V$  (molecules take up space that is unavailable to other molecules).

# The van der Waals Equation: $\left(P + \left(\frac{n}{V}\right)^2 a\right)(V - nb) = nRT$

## Constant $a$

1. The ideal-gas law assumes gas molecules do not attract one another.
2. But *real* molecules exert attractive forces (e.g., dispersion forces, dipole interactions).
3. As a molecule approaches the container wall, nearby molecules pull it back toward the interior of the gas.
4. The molecule  $\therefore$  strikes the wall less forcefully than an ideal-gas molecule would.
5. Since pressure arises from molecular collisions with the wall, the measured pressure is lower than the ideal-gas prediction.
6. Since the measured pressure is lower, we add a corrective term to recover the ideal-equivalent pressure:  $P_{\text{ideal}} = P_{\text{measured}} + \left(\frac{n}{V}\right)^2 a$
7. A larger value of  $a$  means stronger intermolecular attractions, as a larger pressure correction is needed.

The correction grows with  $(n/V)^2$  because attractions are pairwise: doubling the molecular density ( $n/V$ ) doubles the number of molecules and doubles the number of neighbors each molecule interacts with, giving a fourfold increase in the total attractive effect.

### *Pairwise interactions:*

A room twice as crowded doesn't just have twice as many conversations; it has roughly four times as many possible pairs of people. Pairwise interactions grow with the square of the crowd's density.

# The van der Waals Equation: $\left(P + \left(\frac{n}{V}\right)^2 a\right)(V - nb) = nRT$

The ideal-gas law neglects intermolecular attractions.



Real molecules attract each other.



Nearby molecules pull wall-bound molecules inward.



Wall collisions become less forceful.



Measured pressure decreases.



Correct by adding  $\left(\frac{n}{V}\right)^2 a$  to measured pressure.

The correction grows with  $(n/V)^2$  because attractions are pairwise: doubling the molecular density  $(n/V)$  doubles the number of molecules and doubles the number of neighbors each molecule interacts with, giving a fourfold increase in the total attractive effect.

*Pairwise interactions:*

A room twice as crowded doesn't just have twice as many conversations; it has roughly four times as many possible pairs of people. Pairwise interactions grow with the square of the crowd's density.



# The van der Waals Equation: $\left(P + \left(\frac{n}{V}\right)^2 a\right)(V - nb) = nRT$

## Constant $b$

1. The ideal-gas law assumes molecules are point particles with zero volume.
2. But *real* molecules occupy physical space.
3. Part of the container's volume is occupied by the molecules themselves.
4. That occupied volume cannot be used for molecular motion.
5.  $\therefore$  the gas actually moves within a smaller free volume ( $V_{\text{free}} = V_{\text{container}} - V_{\text{occupied}}$ ) than the container's total volume.
6. A larger value of  $b$  means larger molecules, so more of the container volume is unavailable for molecular motion.

Unlike intermolecular attractions, which occur between pairs of molecules, excluded volume is a property of individual molecules.

Imagine trying to move around in a room filled with boxes. The room's spatial dimensions have not changed, but the *usable* space has become smaller due to occupation. The more boxes there are in the room, the less usable space there is. If each box has a volume of  $0.5\text{m}^3$ , then 10 boxes will occupy a space of  $10 \times 0.5\text{m}^3$  and  $n$  boxes a space of  $n \times 0.5\text{m}^3 \therefore V_{\text{occupied}} = nV_{\text{box}}$ .

# The van der Waals Equation: $\left(P + \left(\frac{n}{V}\right)^2 a\right)(V - nb) = nRT$

The ideal-gas law assumes zero molecular volume.



Real molecules occupy space.



Molecules reduce available free volume.



Gas moves through less volume.



Use  $V_{\text{free}} = V - nb$ .

Unlike intermolecular attractions, which occur between pairs of molecules, excluded volume is a property of individual molecules.

Imagine trying to move around in a room filled with boxes. The room's spatial dimensions have not changed, but the *usable* space has become smaller due to occupation. The more boxes there are in the room, the less usable space there is. If each box has a volume of  $0.5\text{m}^3$ , then 10 boxes will occupy a space of  $10 \times 0.5\text{m}^3$  and  $n$  boxes a space of  $n \times 0.5\text{m}^3 \therefore V_{\text{occupied}} = nV_{\text{box}}$ .

## Van der Waals Constants

| Gas            | Formula          | $a$ ( $\text{L}^2 \text{atm mol}^{-2}$ ) | $b$ ( $\text{L mol}^{-1}$ ) |
|----------------|------------------|------------------------------------------|-----------------------------|
| Helium         | He               | 0.034                                    | 0.0237                      |
| Hydrogen       | H <sub>2</sub>   | 0.245                                    | 0.0265                      |
| Nitrogen       | N <sub>2</sub>   | 1.39                                     | 0.0391                      |
| Oxygen         | O <sub>2</sub>   | 1.36                                     | 0.0318                      |
| Carbon dioxide | CO <sub>2</sub>  | 3.59                                     | 0.0427                      |
| Water          | H <sub>2</sub> O | 5.46                                     | 0.0305                      |

- ▶ H<sub>2</sub>O has the largest  $a \Rightarrow$  strongest intermolecular attractions (hydrogen bonding).
- ▶ He has the smallest  $a \Rightarrow$  behaves most like an ideal gas (inert).

## What Determines the van der Waals Parameter $a$ ?

- ▶ Question
  - ▶ Why is  $a$  much larger for  $\text{H}_2\text{O}$  than for He?
- ▶ Evidence
  - ▶  $\text{H}_2\text{O}$  has stronger, longer-range attractive interactions than He.
  - ▶ More negative *potential energy* requires more work to separate molecules.
- ▶ Reasoning
  - ▶  $\therefore$  attractions pull molecules inward before wall collisions, momentum transfer decreases.
  - ▶ Molecular size alone cannot deepen the potential well or reduce pressure this strongly.
  - ▶  $\therefore$  larger  $a$  measures deeper attractive potentials and larger pressure corrections.
- ▶ Takeaway
  - ▶  $a$  reflects the depth of intermolecular attraction.

# Why Is the van der Waals Parameter $b$ Larger Than Molecular Volume?

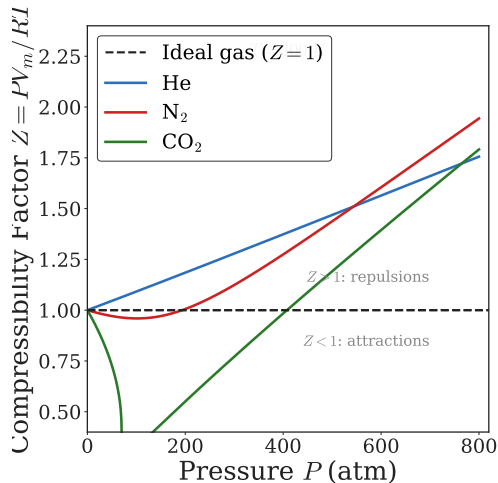
- ▶ Question
  - ▶ Why is  $b$  about four times the volume of a molecule?
- ▶ Evidence
  - ▶ Hard-sphere molecules of radius  $r$  cannot overlap.
  - ▶ A second molecular center is excluded from a sphere of radius  $2r$ .
- ▶ Reasoning
  - ▶  $\therefore V_{\text{excl}} = \frac{4}{3}\pi(2r)^3 = 8\left(\frac{4}{3}\pi r^3\right)$ , excluded volume exceeds molecular volume.
  - ▶ Each excluded region belongs to two molecules, so  $V_{\text{excl}}/2 \approx 4V_{\text{mol}}$ .
  - ▶  $\therefore b$  represents *excluded volume*, not molecular volume.
- ▶ Takeaway
  - ▶  $b$  corrects the volume accessible to molecular centers.

# Real vs. Ideal Gas Behavior

## Compressibility Factor $Z$

$$Z = \frac{PV}{nRT}$$

- ▶ Ideal gas:  $Z = 1$  always.
- ▶  $Z > 1$ : repulsions dominate; gas harder to compress than ideal.
- ▶  $Z < 1$ : attractions dominate; gas easier to compress than ideal.



## Why Does the $Z$ -Factor First Decrease Then Increase?

- ▶ Question
  - ▶ Why does  $Z$  fall below 1 at moderate pressure, then rise above 1?
- ▶ Evidence
  - ▶ Moderate-pressure gases show  $Z < 1$ ; high-pressure gases show  $Z > 1$ .
  - ▶ Attractions lower measured pressure, whereas excluded volume raises it.
- ▶ Reasoning
  - ▶  $\therefore$  attractions dominate first, the  $a$  correction outweighs  $b$ ;  $\therefore Z < 1$ .
  - ▶  $\therefore$  crowding strengthens excluded-volume effects,  $b$  eventually exceeds  $a$ ;  $\therefore Z > 1$ .
- ▶ Takeaway
  - ▶  $Z$  reflects the changing balance between attractions and repulsions.

## Why Does Helium Have $Z > 1$ at All Pressures?

- ▶ Question
  - ▶ Why does helium never show the usual dip below  $Z = 1$ ?
- ▶ Evidence
  - ▶ Measurements show helium has  $Z > 1$  over the entire pressure range.
  - ▶ Other gases usually cross from  $Z < 1$  to  $Z > 1$  as pressure increases.
- ▶ Reasoning
  - ▶  $\therefore$  helium lacks a  $Z < 1$  region, the  $a$  attraction never exceeds the  $b$  excluded-volume effect.
  - ▶  $\therefore$  repulsive effects dominate helium's behavior at all measured pressures.
- ▶ Takeaway
  - ▶ Helium remains repulsion-dominated across the measured pressure range.



# What Is the Boyle Temperature?

- ▶ Question
  - ▶ Why does one temperature make  $Z$  initially remain near 1 as pressure increases?
- ▶ Evidence
  - ▶ At one temperature, the initial  $Z$ -vs- $P$  curve is nearly flat.
  - ▶ At lower temperatures it first falls; at higher temperatures it first rises.
- ▶ Reasoning
  - ▶  $\therefore$  *attractions* lower  $Z$  while *excluded volume* raises  $Z$ , equal effects cancel initially.
  - ▶  $\therefore$  the *Boyle temperature* gives nearly ideal behavior only at low pressure.
- ▶ Takeaway
  - ▶ Boyle temperature marks initial attraction–repulsion balance.

## Check Your Understanding

Which gas deviates **most** from ideal behavior at high  $P$  and low  $T$ ?

(A) He    (B) H<sub>2</sub>    (C) N<sub>2</sub>    (D) CO<sub>2</sub>

Come to lecture to see the answer.

## Check Your Understanding

Under what conditions of  $T$  and  $P$  does a real gas behave most like an ideal gas?

Explain using the KMT postulates.

Come to lecture to see the answer.